

THE DEGREE RISK INDEX · FALL 2026 EDITION · FREE SAMPLER

Are the Skilled Trades Safe From AI? The 2026 Risk Score

The free sampler — the score by track, the six factors in brief, and the honest reasons to think twice. The complete profile goes deeper.

2.5

AI EXPOSURE · ELECTRICIAN — SERVICE & MAINTENANCE
where 10 is most at risk

Re-scored every six months. We publish where we might be wrong.

Quick answer: A service electrician scores **2.5 out of 10** for AI exposure, where 10 is most at risk — one of the lowest scores in this index and far safer than most degrees. But the same skills on a production line or in modular construction score **5.0**. The protection was never "working with your hands." It was working with your hands somewhere unpredictable.

The short answer for parents: for the right kid, this is the strongest bet on the board — and almost nobody considers it seriously. No tuition debt, an on-ramp that is getting easier rather than harder, and lower AI exposure than most university paths. The considerations that actually matter are physical wear over decades and income ceilings outside business ownership, not automation.

Skilled trades AI risk score by setting

Every career in this index is scored 1–10, where **10 is most exposed** to AI. Same six factors, same weights, applied identically to an electrician and a paralegal.

TRADE AND SETTING	2023	2025	FALL 2026	3-YR MOVE	BAND
Electrician — service & maintenance	2.1	2.3	2.5	+0.4	Very low
Plumbing / HVAC — service	2.2	2.4	2.6	+0.4	Very low
Welding & fabrication — site work	2.8	3.0	3.2	+0.4	Low
Construction site trades	2.8	3.1	3.4	+0.6	Low
Production line / prefab / modular	4.0	4.5	5.0	+1.0	Moderate

2023 and 2025 figures are reconstructed using current methodology, not archived from past editions.

How that compares: the median career in this edition scores around 5.5. Entry-level software development scores **8.1**. A business analyst scores **7.6**. Bedside nursing scores **2.8**.

The service electrician's +0.4 over three years is the slowest movement in the entire index. For comparison, entry-level software moved +1.8 over the same period.

Will AI and robots replace tradespeople?

Not in the field, and the reason is not sentiment — it is that robotics is decades behind software and lags furthest where the environment is unpredictable.

AI IS TAKING	IT CAN'T TOUCH
Quoting and estimating	Diagnosing a fault in a sixty-year-old building
Scheduling and dispatch	Running new service through a wall with no drawings
Invoicing and job admin	Working in an attic, a crawlspace, a trench
Code and regulation lookup	Judgment when the job isn't what the customer described
Parts ordering and inventory	Being trusted alone in someone's home

For a self-employed tradesperson, almost everything in the left column is a genuine benefit rather than a threat. The office burden of running a trade business is the part being automated.

Why do the trades score so low? The six factors

How a service electrician rates against each. Ratings are 0–10 on each factor's own terms.

FACTOR	RATING	EFFECT ON RISK
How much of this job can AI already do?	5.0	↑ moderate exposure
How hard will it be to land that first job?	2.0	↓ strong protection
Does it have to be done in person, with your hands?	9.5	↓ strongest protection
Does someone need a human they can trust and hold responsible?	8.5	↓ strong protection
Does the law require a licensed human?	9.0	↓ strong protection
How often does the job hit genuinely new, high-stakes situations?	9.0	↓ strong protection

ONE FACTOR, WORKED THROUGH IN FULL

So you can see what the analysis actually looks like.

Does it have to be done in person, with your hands? — and why "manual work" is the wrong category

This factor is usually described as though it were binary: desk jobs are exposed, manual jobs are safe. That framing is wrong, and the trades are where it breaks most visibly.

What actually matters is not whether the work is physical. It is whether the environment is predictable.

Robotics performs well in controlled settings — known dimensions, repeated tasks, fixed tolerances, no surprises. That is a warehouse, an assembly line, a modular construction facility. It performs badly in uncontrolled ones, and the gap is not narrowing quickly.

A service electrician spends the day in unpredictable environments by definition: a different building every job, unlabeled wiring, work someone else did badly in 1974, a customer describing the problem inaccurately. That rates 9.5.

The same trade skills applied to a production line rate 8.0 — still physical, but in exactly the setting robotics handles best. That single difference, plus lower trust and licensing requirements, is what separates a 2.5 from a 5.0.

So the useful question for any hands-on career is not "is this manual work?" but would two days of this job ever look the same? If the answer is no, the protection is real. If the answer is yes, physical work offers much less shelter than people assume.

Which trades are safest from AI?

Ranked by exposure, safest first:

1. **Electrical service and maintenance** — 2.5. Unpredictable environments, licensed, high trust, high consequence. 2. **Plumbing and HVAC service** — 2.6. Same shape, marginally less regulated in some jurisdictions. 3. **Welding and fabrication on site** — 3.2. Physical and skilled, but with more repeatable work and weaker licensing. 4. **Construction site trades** — 3.4. Genuinely varied work, though modular methods are shifting some of it indoors. 5. **Production line and prefab** — 5.0. Structured, repeatable, and the setting where automation advances fastest.

Is a trade better than a degree in 2026?

For the right person, on the evidence we have, often yes — and this is the comparison most families never actually run.

The trades' on-ramp is *strengthening* while most degree on-ramps narrow. Apprenticeships are paid rather than charged for, legally embedded in licensing, and chronically undersubscribed as the existing workforce ages toward retirement. Analysis of job postings found skilled-trades demand grew 27% over four years, outpacing desk-based professional roles by 19 percentage points — over the same period that unemployment among recent college graduates hit its highest level in more than a decade outside the pandemic.

Electrical work in particular sits directly in front of the electrification and data-center buildout. It is the one trade with a demand tailwind coming from AI itself.

The honest other side: physical wear over thirty years is real, income ceilings outside business ownership are real, and much of the current construction demand is cyclical rather than structural — data-center jobs end when the build does. A trades career entered on the strength of today's wages could meet a very different market in 2032. The demographic case, though, holds regardless of the cycle.

Common questions

Will robots replace electricians? Not in service and repair work. Robotics advances fastest in structured, predictable environments, and no two service calls are alike. Electrical work in new-build and prefab settings is more exposed.

Is a trade a better bet than college? On AI exposure, most trades beat most degrees — a service electrician at 2.5 against a business analyst at 7.6 or an entry-level developer at 8.1. Add no tuition debt and a paid apprenticeship and the financial comparison is strong. The trade-offs are physical and financial rather than technological.

Which trade has the best future? Electrical, on current evidence — it sits in front of grid and data-center demand created by AI itself, and it carries strong licensing protection.

Do the trades pay well? Well enough that the comparison with a mid-tier degree is genuinely close, and business ownership raises the ceiling considerably. Without ownership, income tends to plateau earlier than in professional careers.

Are apprenticeships hard to get? Less hard than most graduate schemes. The sector is short of entrants, and public investment in registered apprenticeships has been expanding.

Related profiles

- **Engineering** — 4.0 licensed and site-based, 6.3 desk and CAD
 - **Construction & project management** — coming in this edition
 - **Computer science** — 8.1 entry-level, 5.4 senior. Free to read in full.
 - **Nursing** — 2.8, the other career with a legally protected on-ramp
 - **Business and management** — 7.6 analyst, 4.9 leadership
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What's in the full trades profile

This sampler tells you where the trades stand. The full profile tells you what to do about it.

	SAMPLER	FULL PROFILE
Verdict, all setting scores, 3-year trend	✓	✓
Six-factor ratings	✓	✓
Reasoning behind every factor rating	one example	✓
How durable each protection is — including where physical AI is heading		✓
The honest downsides — physical wear, income ceilings, cyclical demand		✓
What's genuinely good about it — and who this work suits		✓
The AI-native advantage — how to prepare, and why this cohort has an opening		✓
Routes in — union versus open-shop apprenticeships, licensing timelines		✓
Where it leads later — ownership, contracting, specialization		✓
How to evaluate an apprenticeship		✓
Questions to ask a program or employer — plus red flags		✓
Sourced further reading, including the case against our score		✓
Discussion questions for parent and student		✓
A short version written directly to the student		✓
Technical scoring appendix		✓

GET THE FULL PROFILES

Most families are weighing two or three careers seriously, and a few more they haven't ruled out. Pick the ones you need.

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Each full profile includes everything in the table above, plus a short version written directly to the student and the technical scoring appendix. Spring 2027 updates of whatever you buy are included.

[Read a complete profile free →](#) We publish one profile in full, openly, so you can judge the depth before buying anything. Computer science is the one to read — it's the most surprising result in the index.

Pivotum scores thirty careers against AI exposure using a fixed, published methodology, re-scored every six months. Scores measure exposure to what AI can already do — not how much any particular employer has deployed. We publish where we might be wrong.